

Custom ServiceNow Application: Library Services Management

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Scope: `u_borrow_request_mani` / borrow request 1

System Lifecycle State: 100% Configured, Automated, and Verified

1. Core Architecture & Schema

Data Tables & Field Configuration

- **Book Table (`u_book`)**
 - Tracks the available library inventory.
 - **Key Fields:** Title, Author, ISBN, Status (*Available, Borrowed, Maintenance*).
 - **Reference Qualifier:** Configured to dynamically filter and display only books where `Status = Available` when creating a new request.
 - **Borrow Request Table (`u_borrow_request_mani`)**
 - Handles the lifecycle of a student transaction.
 - **Form Variables & Behaviors:**
 - **Book Title & Author:** Reference field to the Book table.
 - **Borrower:** Auto-populates with the active session user; remains editable for librarians borrowing on behalf of others.
 - **Borrow Date:** Auto-filled with the system current date.
 - **Return Date:** Set via UI Policy to be hidden/optional initially, becoming **mandatory** only when the request status changes to `Issued`.
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2. Process Automation (Flow Designer)

The end-to-end execution path maps directly to the following state machine tracking:

`Requested` $\rightarrow$ `Approved` $\rightarrow$ `Issued` $\rightarrow$ `Returned`

Execution Steps & Use Cases

1. **Trigger:** A new record is inserted into the Borrow Request table.
2. **Approval Logic:**
 - *Student Submissions:* Automatically routes an approval record to the **Librarian Group**.
 - *Librarian Submissions:* Triggers an auto-approval rule for internal/testing bypass.
3. **Fulfillment Tasking:** Upon approval, the flow creates a **Catalog Task (`sc_task`)** assigned to the Librarian group to physically locate and hand over the book.
4. **Status Syncing:**
 - Moving the task/process forward shifts the Borrow Request status to **Issued** (locking in the Return Date UI constraint).

- Marking the book as physically returned updates the record status to **Returned** and flags the underlying Book asset back to **Available**.
- 5. **Notifications:** Automated email notification triggers are configured to alert students immediately when their request is **Approved** and when the book is officially **Issued**.

3. Security, Governance, & Reporting

Access Control Lists (ACLs)

- **Student Role:** Restricted to **Create** and **Read** permissions on their own records.
- **Librarian Role:** Full **Read**, **Write**, and **Create** administrative permissions over all records across both tables.

Reporting Engine

- **Report Name:** "Most Borrowed Books"
- **Visualization:** Configured via the [Report Designer View](#) to surface active bottlenecks and historical usage patterns.
- **Active Criteria Filter:** Strictly isolated to **status | is | approved** to isolate successfully processed transactions.

4. Portal User Experience Walkthrough

Navigation Path

[Home](#) → [Service Catalog](#) → [Library Services](#) → [Submit Borrow Request](#)

Role-Based Narration Stream

- **The Student Perspective:** *"A student accesses the user-facing Service Portal, navigates to Library Services, and searches for an available title. They populate the form and hit submit. The session context auto-fills their credentials, generating an immediate backend record with a state of **Requested**."*
- **The Librarian Perspective:** *"The librarian receives an automated notification dashboard widget item. They review the timeline, issue the approval, clear the physical fulfillment task, and hand over the asset. The platform dynamically handles the state transitions and mandates the return metrics until final drop-off."*